C PROGRAMMING

Lesson 9 — Nested If Statements

Based on: Bro Code C Tutorial Series

1. What is a Nested If Statement?

A nested if statement is simply an if statement placed inside another if statement. You use this when a decision depends on multiple conditions that are related to each other in a hierarchy.

```
if (outer condition) {
    if (inner condition) {
        // both conditions are true
    } else {
        // only outer is true
    }
} else {
    // outer condition is false
}
```

> **Key Idea:** The inner if only runs if the outer if is already true. This lets you make decisions that depend on multiple related conditions.

2. Why Nesting Matters — The Discount Problem

This lesson uses a movie ticket program to show exactly why nesting is necessary. The base price is $10.00 with these discounts:

- **Student only** → 10% off → $9.00
- **Senior only** → 20% off → $8.00
- **Both** → 30% off → $7.00
- **Neither** → no discount → $10.00

The Wrong Way — Sequential discounts

If you apply discounts one after another without nesting, both discounts stack incorrectly:

```
// WRONG approach
if (is_student) {
    price *= 0.9;   // 10% off --> $9.00
}
if (is_senior) {
    price *= 0.8;   // 20% off $9.00 --> $7.20  WRONG!
}
```

```
// For student + senior: result is $7.20, not $7.00
```

> **The Bug:** Two separate if statements both run independently. So for student + senior, 10% is taken off first, then 20% is taken off the already-reduced price. That gives $7.20 instead of the correct $7.00.

The Right Way — Nested if statements

Nesting lets you handle all combinations correctly in one logical structure:

```
if (is_student) {
    if (is_senior) {
        price *= 0.7;   // student + senior: 30% off --> $7.00
    } else {
        price *= 0.9;   // student only: 10% off --> $9.00
    }
} else {
    if (is_senior) {
        price *= 0.8;   // senior only: 20% off --> $8.00
    }
    // else: no discount, price stays $10.00
}
```

3. Practice Problem

Practice: Gym Membership Pricing

Write a C program that calculates a gym membership price based on membership type and whether the person is a student.

Rules:

- Base price: $50.00 per month
- Premium member gets 20% off
- Student gets an additional 10% off (applied on top of any existing discount)
- Non-premium, non-student pays full price

Use bool variables: is_premium and is_student

Expected Outcomes

is_premium	is_student	Final Price
true	true	$36.00
true	false	$40.00
false	true	$45.00
false	false	$50.00

> **Hint:** Unlike the movie ticket example, here the student discount stacks ON TOP of the premium discount — so apply premium first, then student. Think carefully about where to nest!